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Introduction to Networking (032025-NIO)

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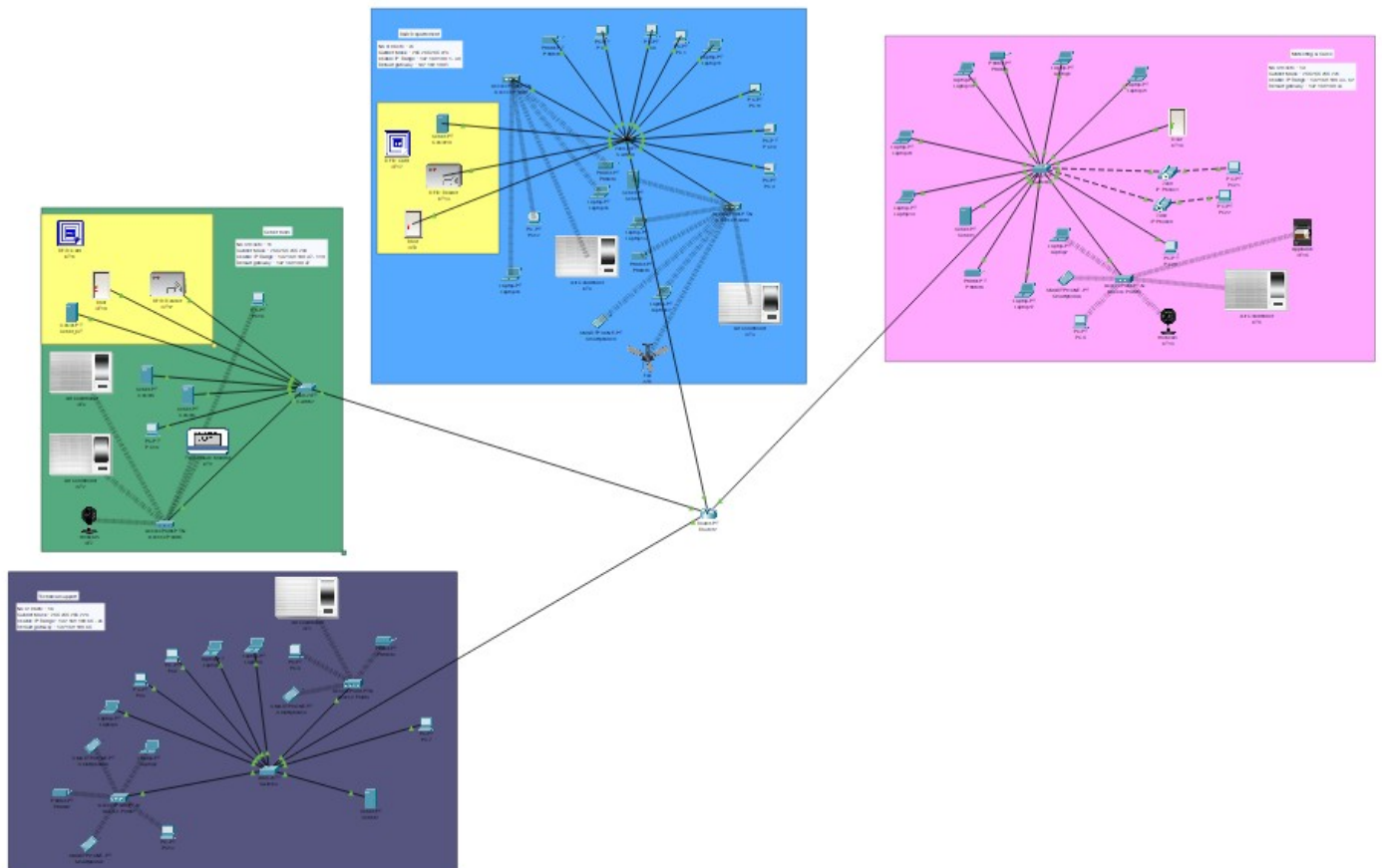


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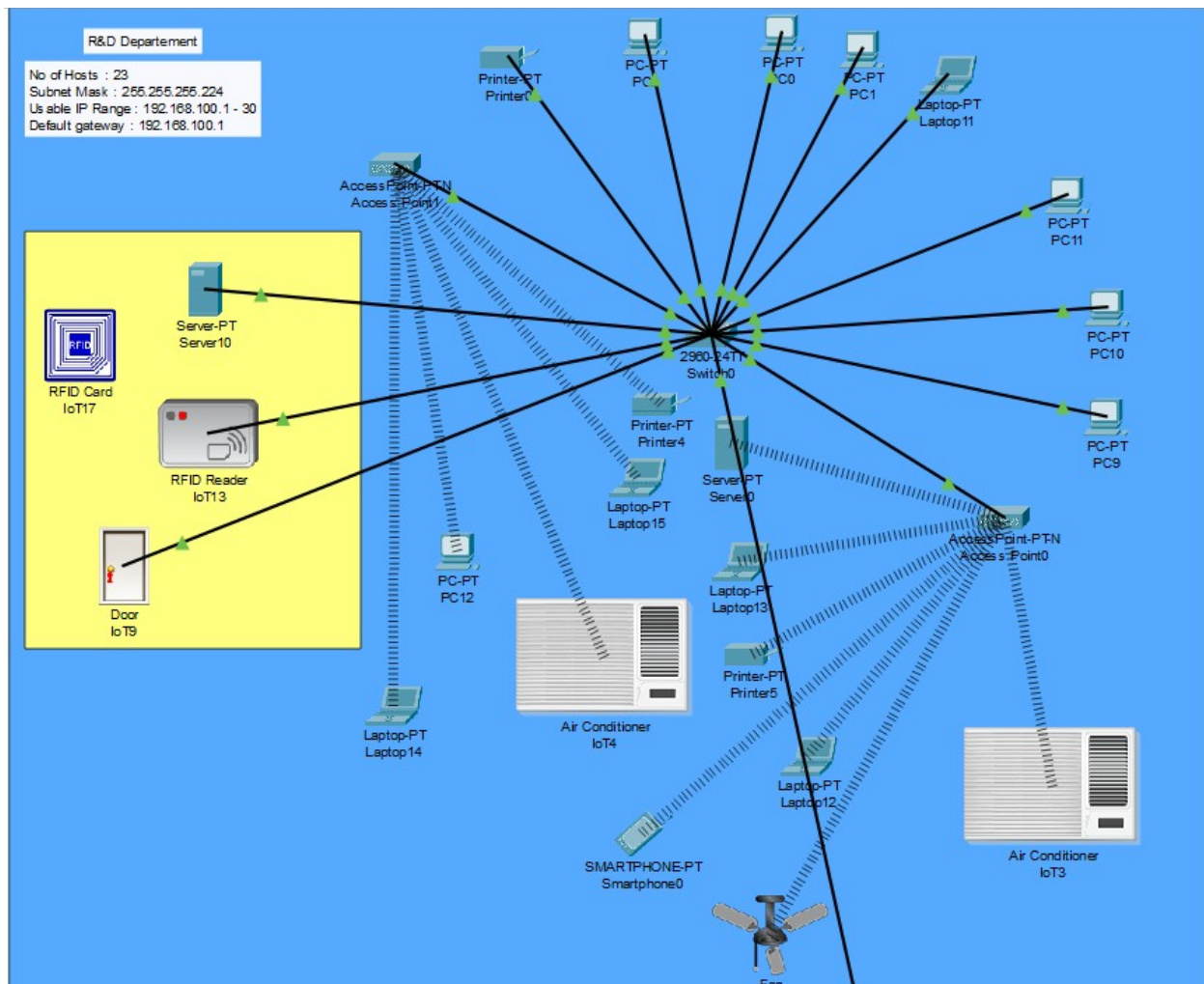
1.0 Introduction :

In the modern-day world, where networks are built wherever they are needed, it is essential to design the networks effectively when it comes to ensuring the smooth flow of communication, scalability and security among different departments of an organization. The present report provides a broad-based subnetting plan to a multi departmental enterprise network covering the best utilization of IP address space, connection of devices and up-scale development.

The network itself has been divided into the main departments: Researches and Development (R&D), Marketing and Sales, Technical support, Server Room, with each host having specific host requirements. In order to make the address as efficient as possible, Variable-Length Subnet Masking (VLSM) is used in the report, enabling use of subnet sizes that fit specific requirements and grow with them at the same time wasting as little IP as possible. Also, to facilitate a higher level of security and automate the working process, IoT gadgets (RFID accessibility, automatic doors, and environmental sensors) are deployed in the design.



2.0 R&D Department :



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R&D Departement

No of Hosts : 23

Subnet Mask : 255.255.255.224

Usable IP Range : 192.168.100.1 - 30

Default gateway : 192.168.100.1

The Reason to Use this Subnet Mask:

Host Requirements: 23 hosts are available in R&D department. A 255.255.255.224 (/27) subnet mask supplies:

The complete IP numbers are 32 (with 30 usable sum, because the first one should be allocated to the network ID, and the last to the broadcast).

Adequate Size: This will support the 23 hosts leaving some expansion space (7 additional addresses).

Effective Addressing: The /27 subnet is convenient, and recommended in small to medium size networks as a compromise with address conservation and ease of use. Severe wastage of addresses would occur in larger subnets (e.g. /26), whereas smaller ones (e.g. /28) would not be enough.

Default Gateway: The first available IP (which is 192.168.100.1) has been configured as the gateway, as is usual to have consistency.

Configure from the router :

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.100.1 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
```

So What is the Rationale of This Configuration?

FastEthernet 0/0 interface in the router would be set-up as a default gateway (192.168.100.1) in the R&D subnet.

The subnet mask (255.255.255.224) will provide 30 usable hosts fitting within the R&D department with 23 hosts, which leave provisions, given the expansion possibility.

No shutdown is absolutely important otherwise the interface would be still disabled and devices would not be able to communicate via the gateway.

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Configure from the switch :

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.100.2 255.255.255.224
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to
up

Switch(config-if)#exit
```

Its configuration starts with entering global configuration mode after which the interface number 1 in the switch is found in the VLAN 1 as the default management VLAN along which IP address is assigned. VLAN 1 is usually associated with simple configurations, but in contemporary security best practice management traffic should be segregated into its own VLAN to reduce risk. However, the VLAN 1 is assigned the IP address as 192.168.100.2 with the subnet mask of 255.255.255.224 (/27) that puts it on the same subnet as the gateway of the router (192.168.100.1). Such synchronization will establish smooth communication between the switch and other equipments in the R&D network.

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip default-gateway 192.168.100.1
Switch(config)#exit
Switch#
```

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These steps will be initiated with the enable command, which will enable us to access the privileged EXEC mode, and then by entering in the global configuration mode, configure terminal. In this case; the administrator would like to set the default gateway to be 192.168.100.1 using option of ip default-gateway 192.168.100.1. This command will represent the interface (192.168.100.1) of the router as the gateway of the switch that will enable it to redirect data to other networks when needed.

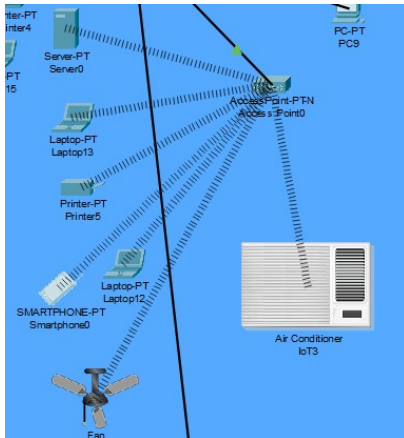
The configurations for the end devices (IP – Subnet mask – Default gateway) :

The diagram shows a central router with six end devices connected to it. Each device has a configuration panel with the following settings:

- Device 1: Static IP 192.168.100.25, Subnet Mask 255.255.255.224, Default Gateway 192.168.100.1
- Device 2: Static IP 192.168.100.24, Subnet Mask 255.255.255.224, Default Gateway 192.168.100.1
- Device 3: Static IP 192.168.100.23, Subnet Mask 255.255.255.224, Default Gateway 192.168.100.1
- Device 4: Static IP 192.168.100.22, Subnet Mask 255.255.255.224, Default Gateway 192.168.100.1
- Device 5: Static IP 192.168.100.21, Subnet Mask 255.255.255.224, Default Gateway 192.168.100.1
- Device 6: Static IP 192.168.100.20, Subnet Mask 255.255.255.224, Default Gateway 192.168.100.1

Below the router, there is a configuration panel for the router itself:

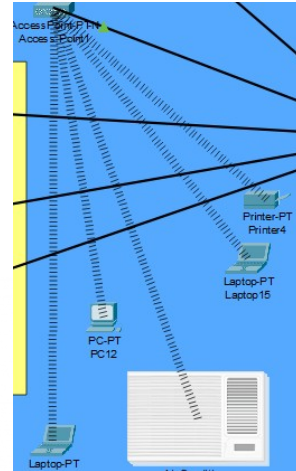
- Static IP Address: 192.168.100.18
- Subnet Mask: 255.255.255.224
- Default Gateway: 192.168.100.1



IP address : 192.168.100.3 - 9

Subnet Mask :
255.255.255.240

Default gateway : 192.168.100.1

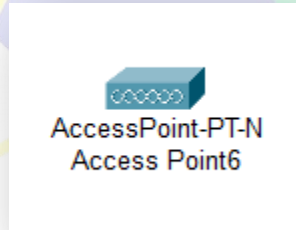


IP address : 192.168.100.9 - 13

Subnet Mask :
255.255.255.240

Default gateway : 192.168.100.1

How to create access point :

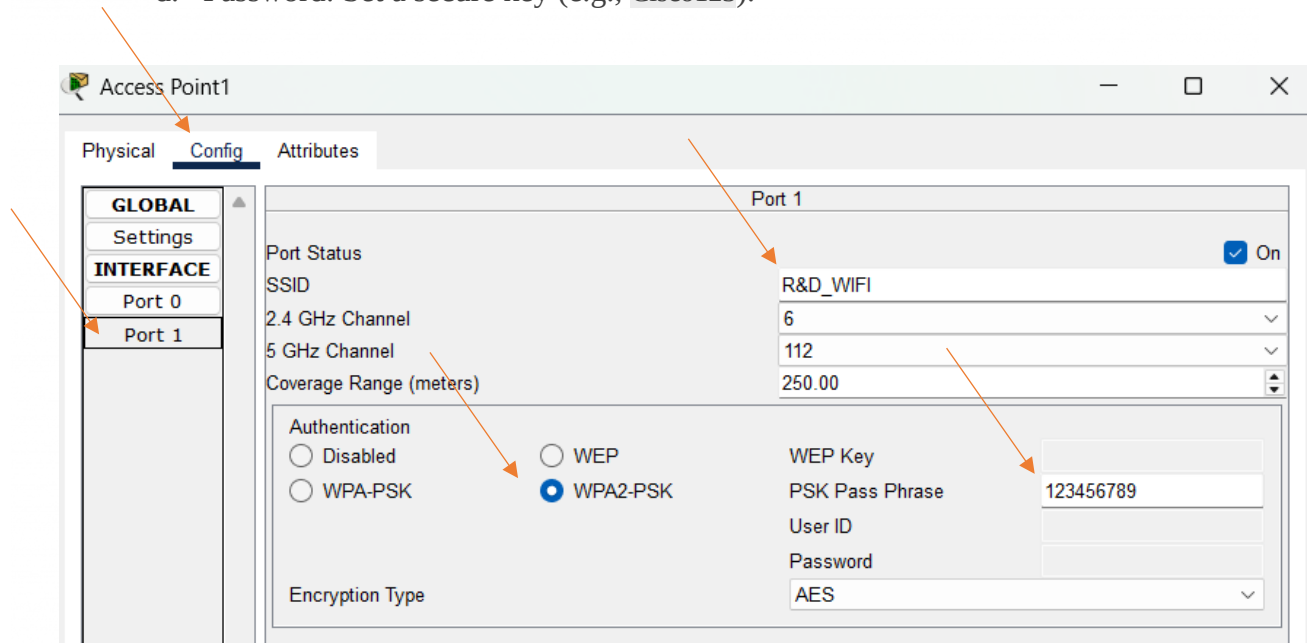


1- add a Wireless Router

Click Wireless, Select > go to Devices Wireless Select e.g. Wireless Router0 (Wireless-PT).
Drag it to the working area.

2- Configure Wireless Settings

- a. Go to config then select port 1.
- b. SSID: Set a name (e.g., MyWiFi).
- c. Authentication: WPA2-PSK (recommended).
- d. Password: Set a secure key (e.g., Cisco123).



How to connect with wireless :

- 1- Open end device then go to desktop



- 2- Then click on PC wireless

- 3- Search and select the wifi

Link Information

Connect

Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
R&D_WIFI_2	1	63%
Marketing & Sales	1	63%
Technical_support...	1	63%
Default	1	63%

Refresh

Connect

Site Information

Wireless Mode

Infrastructure

Network Type

Mixed B/G/N

Radio Band

Auto

Security

WPA2-PSK

MAC Address

0050.0F39.7676

WPA2-Personal Needed for Connection

This wireless network has WPA2-Personal enabled. To connect to this network, enter the required passphrase in the appropriate field below. Then click the **Connect** button.

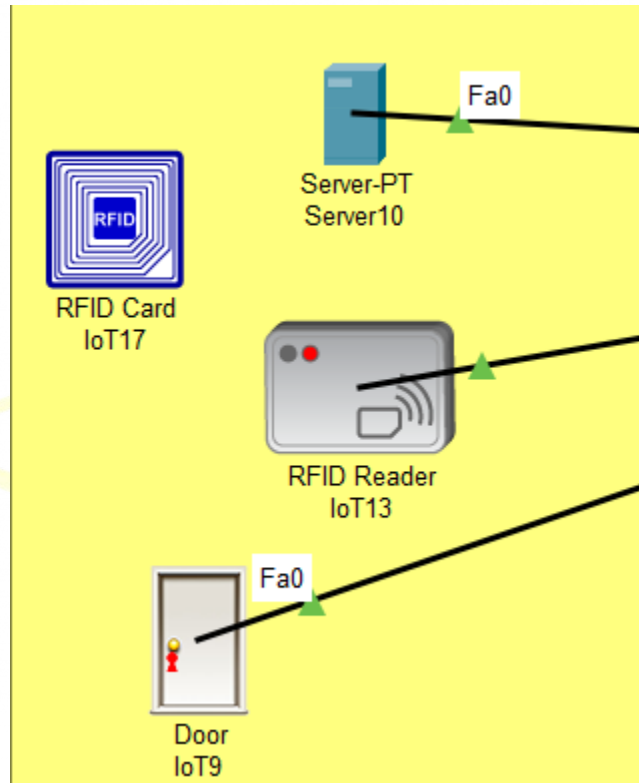
Security

WPA2-Personal

Please select the wireless security method used by your existing wireless network.

Pre-shared Key

Please enter a Pre-shared Key that is 8 to 63 characters in length.



RFID Card (IoT17) and RFID Reader (IoT13)

Creates an RFID based access control system where the value of the card will be scanned by the reader.

Pros: Allows the use of contact-less buildings or restricted areas entrance systems.

Server-PT (Server10)

Performs the role of the central management system, so it is probably processing data of the RFID system and IoT devices.

Pro: Stores, processes, and controls policies of data access into a centralized place enabling efficient management.

Door (IoT9)

Stands for an automated door operated with the help of the RFID system.

Pro: It is more secure since only verified individuals with the appropriate security clearance (RFID validation) can access the possibly restricted area.

1- configure form Server-PT (Server10) :

Server10 1-

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP 2-
- IoT**
- VM Management
- Radius EAP

This service runs on top of the HTTP or HTTPS service.

Registration Server ☒ On 3-

	Username	Password
1	admin	admin

2- Configure from the door (IoT9) and RFID Reader (IoT13) :

IoT Server

☐ None

☐ Home Gateway

☒ Remote Server

Server Address

User Name

Password

IoT Server

☐ None

☐ Home Gateway

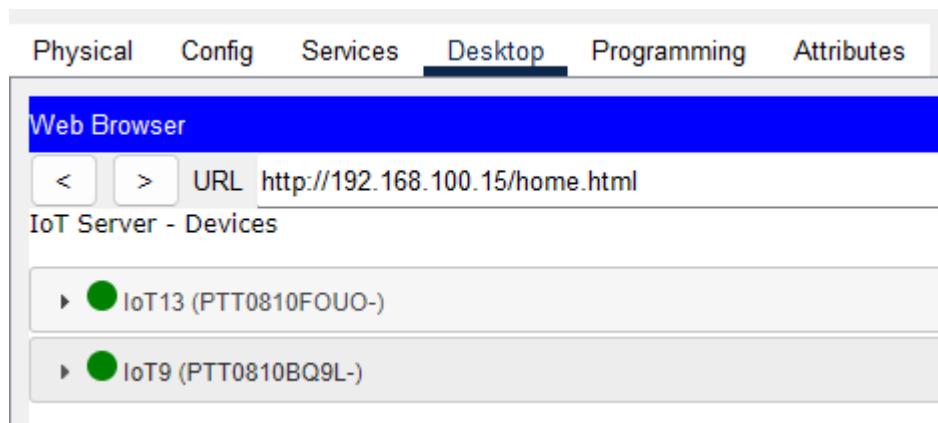
☒ Remote Server

Server Address

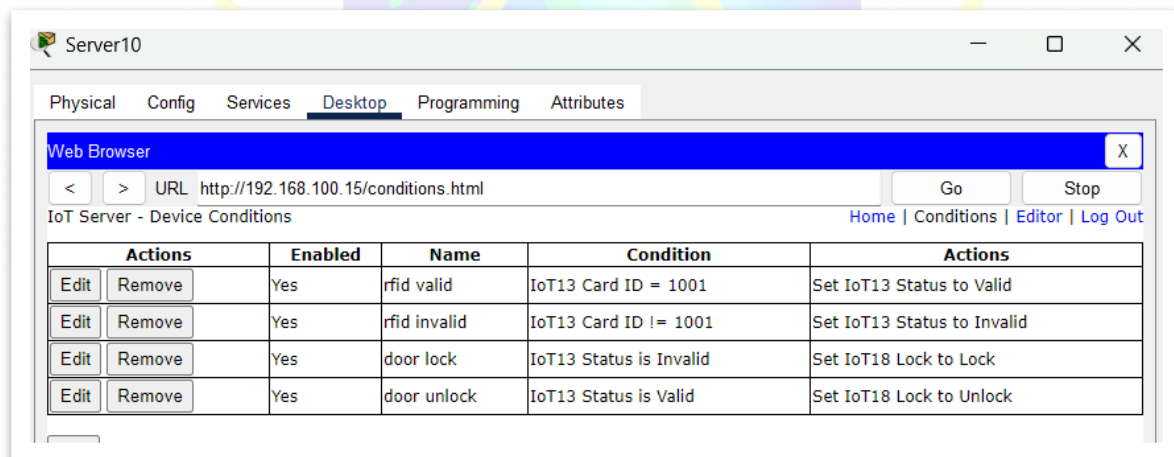
User Name

Password

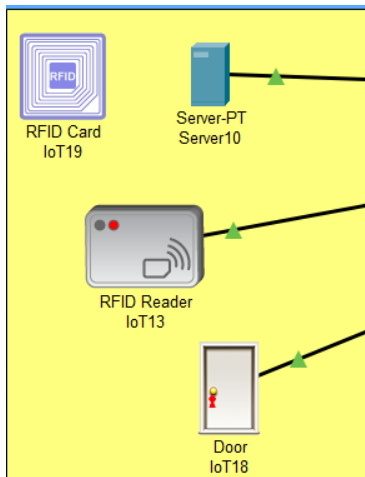
3. Check the end devices are connected to the server :



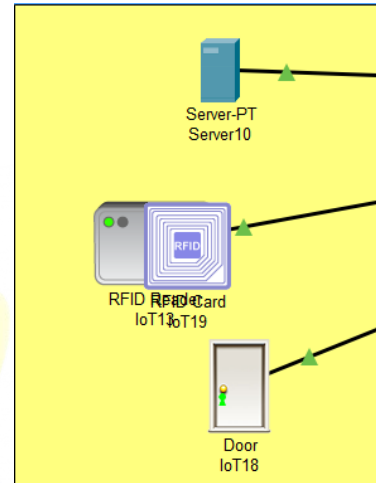
4- Add the conditions



5- Check the conditions if it works :



-- Before --

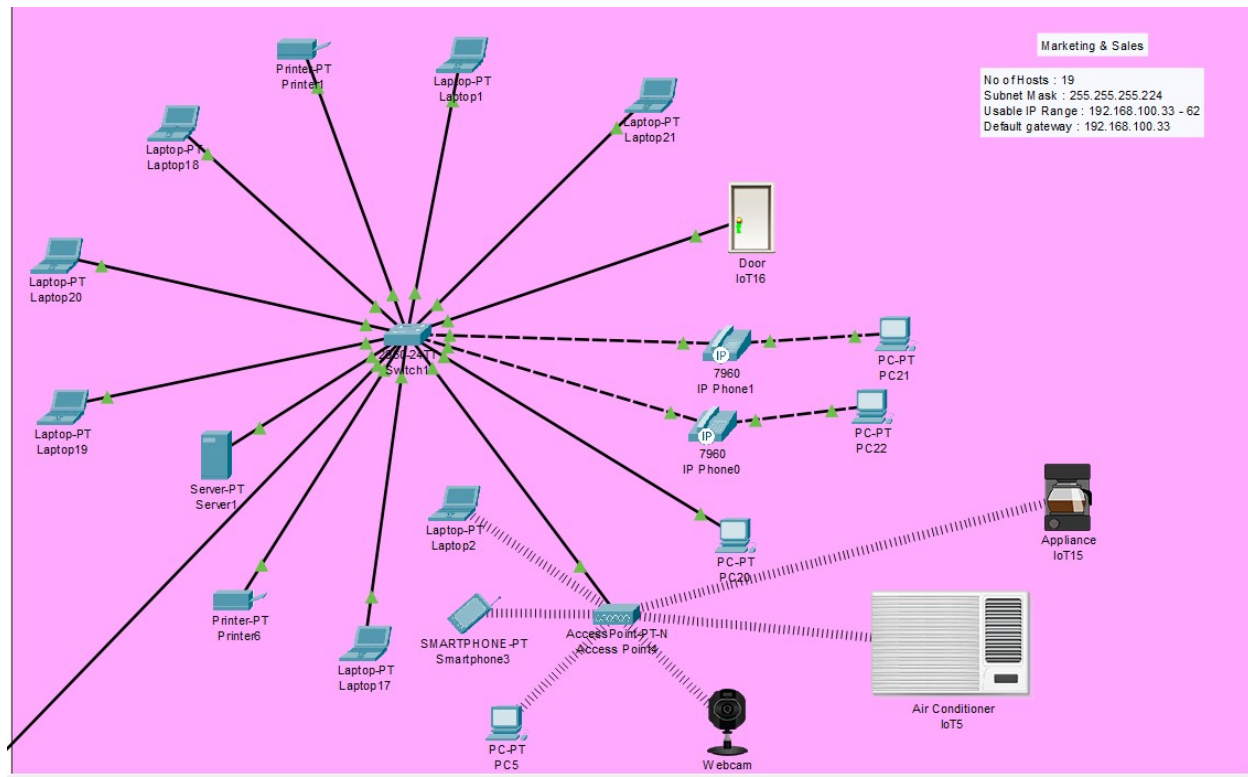


-- After --

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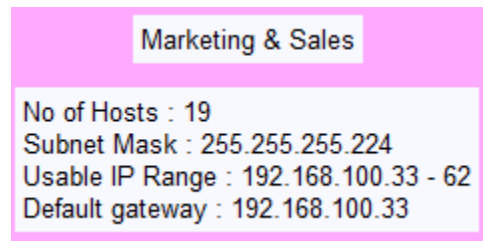
3.0 Marketing & Sales Department:



Subnet Design: Our department uses the helper subnet 192.168.100.32/27 (mask: 255.255.255.224), and offers 30 useful IP addresses (33 62) to host 19 systems and has a remaining capability of expansion.

The routing interface is default gateway: 192.168.100.33 through which it can communicate to other network segments.

Connected Devices: Between workstations (laptops, PCs), servers, smartphones, and IoT (e.g., "All Conditioner IoT5") are used in the department--this is nearly an ideal combination of connectivity needs in the department.



- 1- Scalability: 11 free assignable addresses to support extra devices without reconfiguration

- 2- Isolation: Isolated subnet More security and traffic control offense
- 3- Uniformity: Inconsistency /27 subnetting of departments makes administration easy
- 4- Optimization: No cognizant addresses forfeited (relative to the utilization of a bigger /26 subnet)

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet7/0
Router(config-if)#ip address 192.168.100.33 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

The example cited in this configuration snippet is establishment of a GigabitEthernet interface on a Cisco router, which is a primary activity during implementation of networks. The commands used here are the commands which assign an IP to the interface and displays the interface to be in operation so that it is able to engage in network communication.

Network Segmentation: The /27 subnet is a good physical division of the network as it accepts medium-sized network segments such as server farms or departmental LANs.

Speedy Connectivity: The GigabitEthernet guarantees quick transmission of data, appropriate in high bandwidth activities.

Operation Preparedness - The no shutdown should be entered vital - Interfaces are defaulted to administratively down and will not operate until activated.

Configure the Switch (VLAN 1 + Default Gateway)

```
Marketing&sales>en
Marketing&sales#confi
Marketing&sales#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Marketing&sales(config)#interface v
Marketing&sales(config)#interface vlan 1
Marketing&sales(config-if)#ip address 192.168.100.34 255.255.255.224
Marketing&sales(config-if)#no sh
Marketing&sales(config-if)#no shutdown
Marketing&sales(config-if)#exit
Marketing&sales(config)#
```

Configuration of VLAN Interface VLAN

VLAN one is set up with an IP of 192.168.100.34/27

This makes it on the same subnet as earlier devices (192.168.100.32/27 network)

Subnet Mask 255.255.255.224 gives 30 Available host addresses

Set the default gateway (router's IP)

```
Marketing&sales#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Marketing&sales(config)#interface vlan1
Marketing&sales(config-if)#ip default-gateway 192.168.100.33
Marketing&sales(config)#exit
Marketing&sales#
%SYS-5-CONFIG_I: Configured from console by console
```

The configurations for the end devices (IP – Subnet mask – Default gateway) :

☒ Static	
IPv4 Address	192.168.100.48
Subnet Mask	255.255.255.224
	192.168.100.33
	0.0.0.0

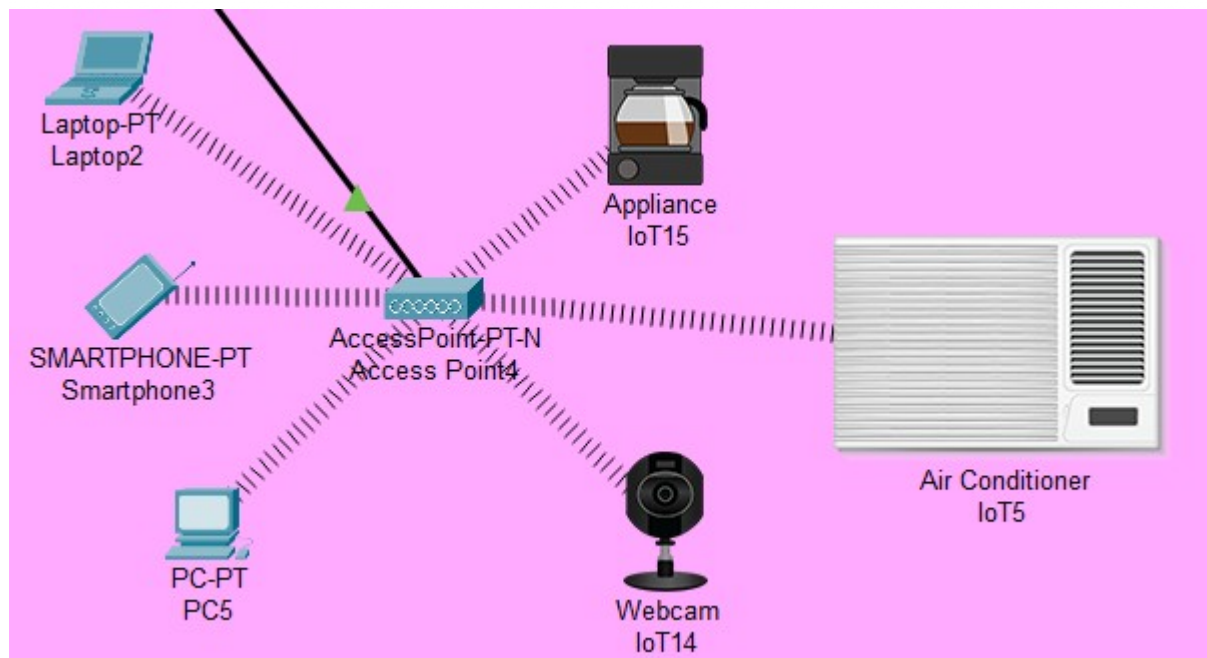
☒ Static	☒ Static	☒ Static
192.168.100.50	192.168.100.52	192.168.100.53
255.255.255.224	255.255.255.224	255.255.255.224
192.168.100.33	192.168.100.33	192.168.100.33
0.0.0.0	0.0.0.0	0.0.0.0

☒ Static	
IPv4 Address	192.168.100.51
Subnet Mask	255.255.255.224
	192.168.100.33
	0.0.0.0

Default Gateway 192.168.100.33	
--------------------------------	--

☒ Static	☒ Static
192.168.100.36	192.168.100.38
255.255.255.224	255.255.255.224
192.168.100.33	192.168.100.33
0.0.0.0	0.0.0.0

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.100.37
Subnet Mask	255.255.255.224



IP address : 192.168.100.39
- 44

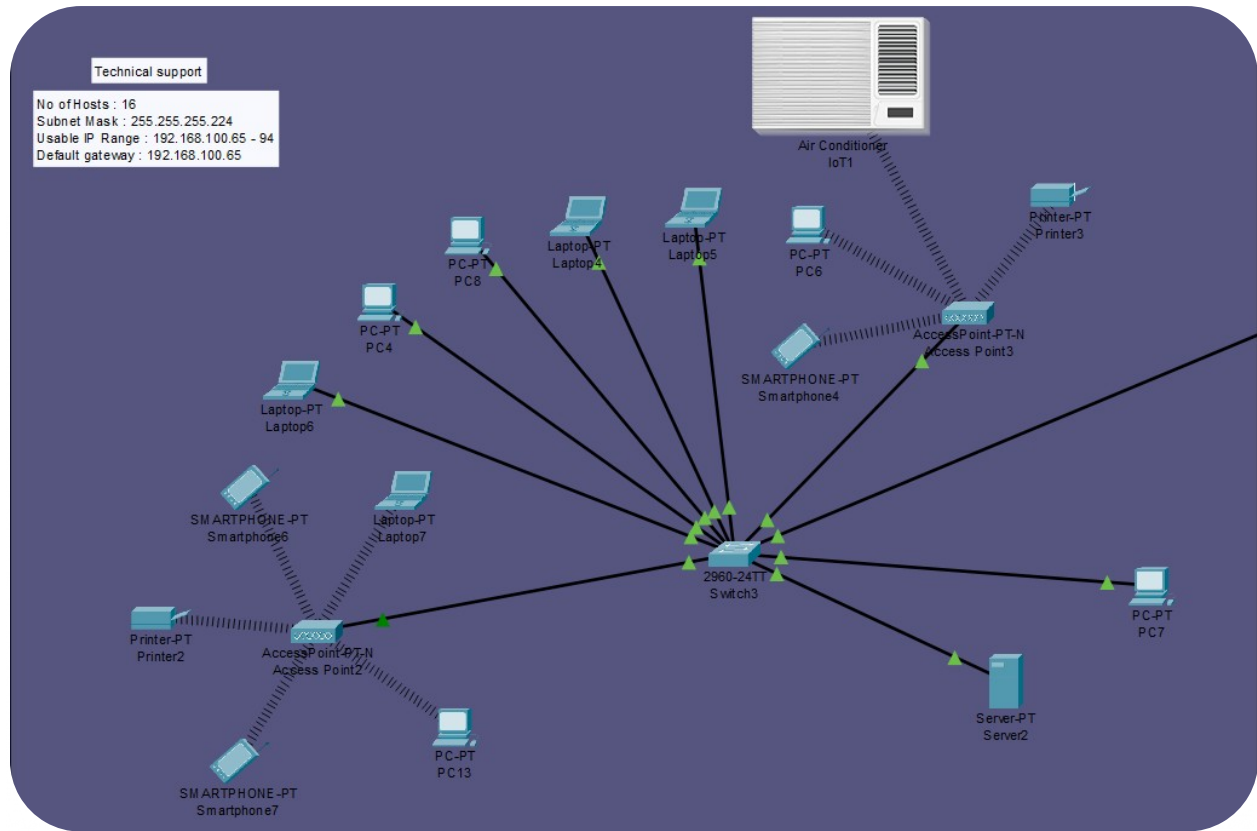
Subnet mask :
255.255.255.224

Default - gateway :
192.168.100.33

☒ Static ☐ DHCP	
192.168.100.46	192.168.100.47
255.255.255.224	255.255.255.224
192.168.100.33	192.168.100.33
0.0.0.0	0.0.0.0

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.100.48
Subnet Mask	255.255.255.224

4.0 Technical Support Department:

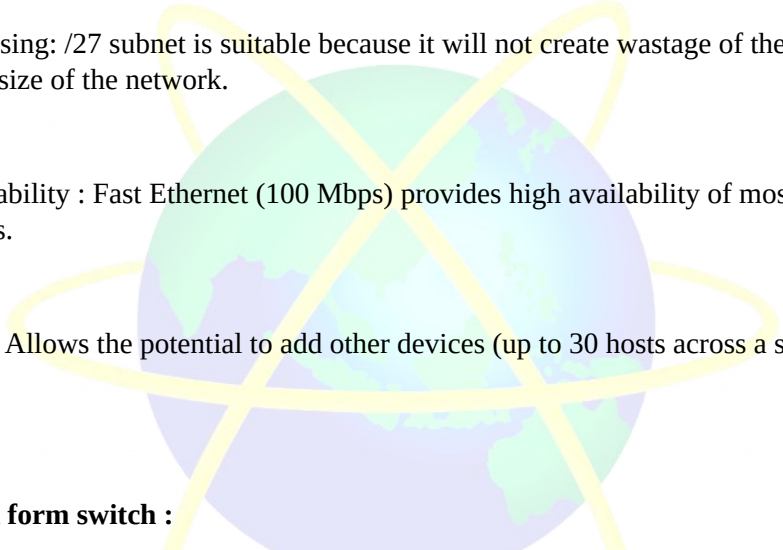


This design describes the backbone network of a Technical Support division, including a well-planned IP addressing system and attached devices that support the operations of the employees as well as IoT-enhanced environmental controls.

- **IP Allocation:**

- o Usable Range: 192.168.100.65 - 94
- o Gateway: 192.168.100.65 (router interface)
- o Reserved Addresses: 192.168.100.64 (network), 192.168.100.95 (broadcast)

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 1/0
Router(config-if)#ip address 192.168.100.65 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
```

- 
- 1- Best addressing: /27 subnet is suitable because it will not create wastage of the IP addressing against the size of the network.
 - 2- High Availability : Fast Ethernet (100 Mbps) provides high availability of most LAN applications.
 - 3- Scalability: Allows the potential to add other devices (up to 30 hosts across a subnet).

Technical Support form switch :

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.100.66 255.255.255.224
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Meaning and Usefulness :

Remote Management: The IP provides SSH/telnet access to switch.

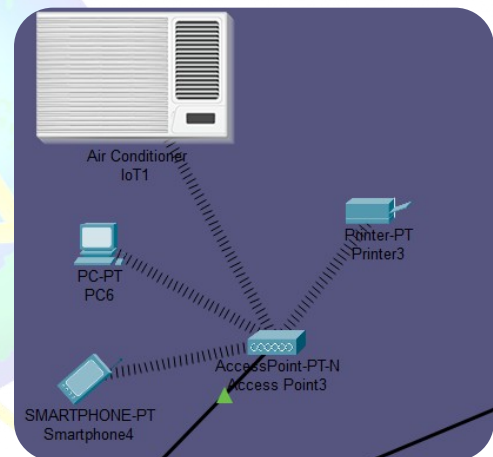
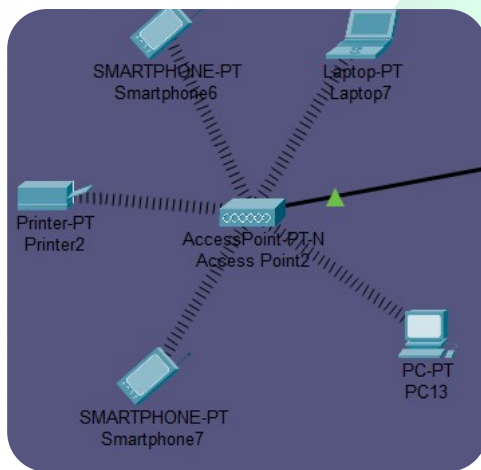
Inter-VLAN Routing: In case the switch is a Layer 3 capable switch, then this SVI can forward traffic between VLANs.

Subnet Consistency: The subnet of the device is consistent with that of other devices (i.e. router of 192.168.100.65).

Set the default gateway (router's IP) :

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip default-gateway 192.168.100.65
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

The configurations for the end devices (IP – Subnet mask – Default gateway) :



IP address : 192.168.100.78

- 82

Subnet mask :
255.255.255.224

Default - gateway :
192.168.100.65

IP address : 192.168.100.69

- 72

Subnet mask :
255.255.255.224

Default - gateway :
192.168.100.65

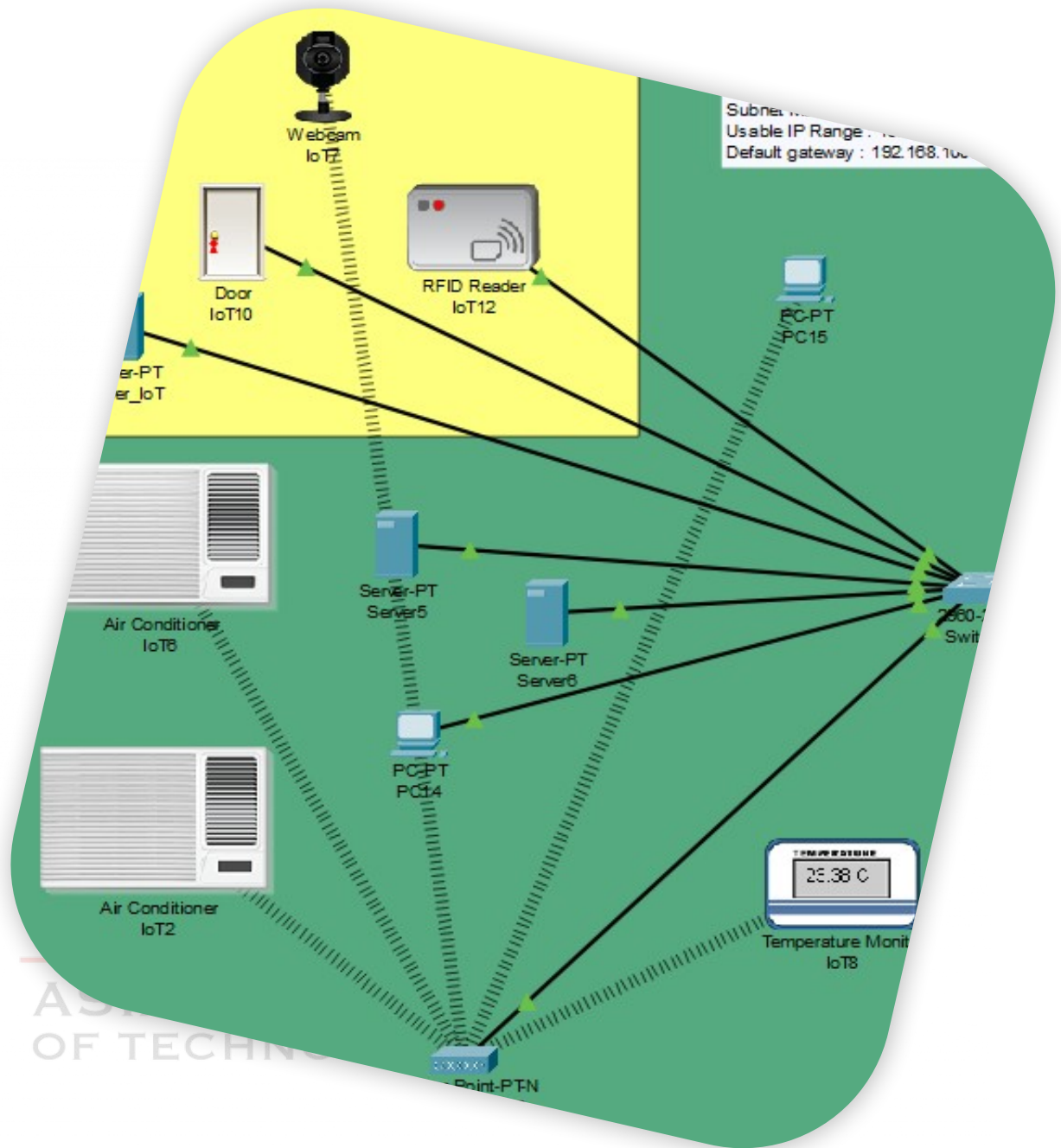
☒ Static	☒ Static	☒ Static
192.168.100.73	192.168.100.74	192.168.100.75
255.255.255.224	255.255.255.224	255.255.255.224
192.168.100.65	192.168.100.65	192.168.100.65
0.0.0.0	0.0.0.0	0.0.0.0

☒ Static	☒ Static	☒ Static
192.168.100.76	192.168.100.77	192.168.100.67
255.255.255.224	255.255.255.224	255.255.255.224
192.168.100.65	192.168.100.65	192.168.100.65
0.0.0.0	0.0.0.0	0.0.0.0

☒ Static
192.168.100.68
255.255.255.224
192.168.100.65
0.0.0.0

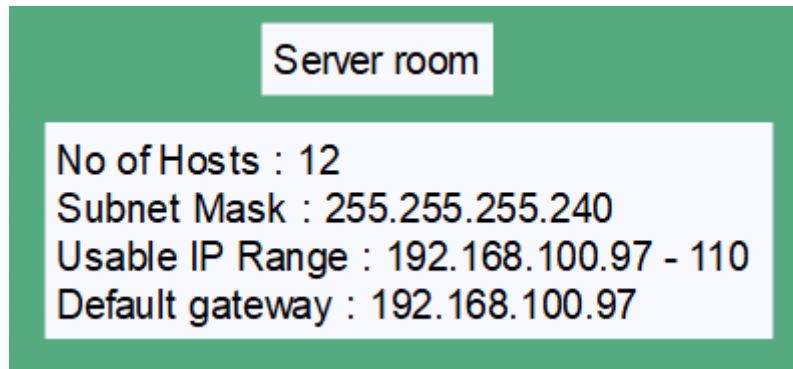
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5.0 Server Room Department :



The design introduces a well-balance yet modernistic server room environment with a strategic integration of the classical network infrastructure and IoT-connected devices to develop a safe and regulated climate-controlled monitoring and carefully controlled technical area. All these are displayed in

the network architecture where the new trend of server rooms can provide the ideal operating situations with the provision of security to the hardware physically.



1. Addressing Scheme

- **Subnet:** 192.168.100.96/28
- **Subnet Mask:** 255.255.255.240
- **Total Addresses:** 16 (14 usable)
- **Current Utilization:** 12 hosts (86% capacity)
- **IP Allocation:**
 - o Network Address: 192.168.100.96
 - o Usable Host Range: 192.168.100.97-110
 - o Gateway: 192.168.100.97 (router interface)
 - o Broadcast: 192.168.100.111

Server Room from the router

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 6/0
Router(config-if)#ip address 192.168.100.97
% Incomplete command.
Router(config-if)#ip address 192.168.100.97 255.255.255.240
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Configuration Breakdown

1. Initial Setup :

- User goes to user EXEC (Router>) mode to privileged EXEC (Router#) mode
- Gets into the global configuration (config t) mode

2. Interface Selection :

- Chooses FastEthernet6/0 interface
- Trenching is the proper selection of a server room connection (wired reliability)

3. Distribution of IPs Process :

- First try (ip address 192.168.100.97) ends in error "% Incomplete command"
- Fixed by introducing subnet mask (255.255.255.240)
- Final issue: ip address 192.168.100.97 255.255.255.240

Server Room form switch :

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.100.98 255.255.255.240
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

1. Interface Configuration :

Switch virtual interface (SVI) VLAN 1:

- IP: 192.168.100.98
- Subnet Mask 255.255.255.240 (/28)
- Second Useable IP in 192.168.100.96/ 28 Subnet
- Router gateway (192.168.100.97) router adjacency

2. Operational Status

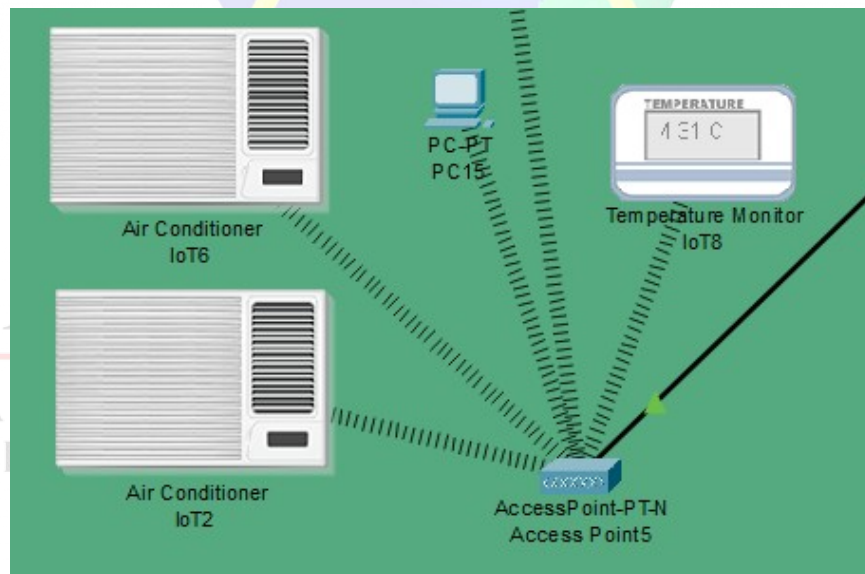
Interface enabled by non shutdown

- Automatic acknowledgement of status of link/line protocol changes
- There are no error messages that show successful implementation

Set the default gateway (router's IP) :

```
Switch(config)#interface vlan1
Switch(config-if)#ip default-gateway 192.168.100.97
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

The configurations for the end devices (IP – Subnet mask – Default gateway) :

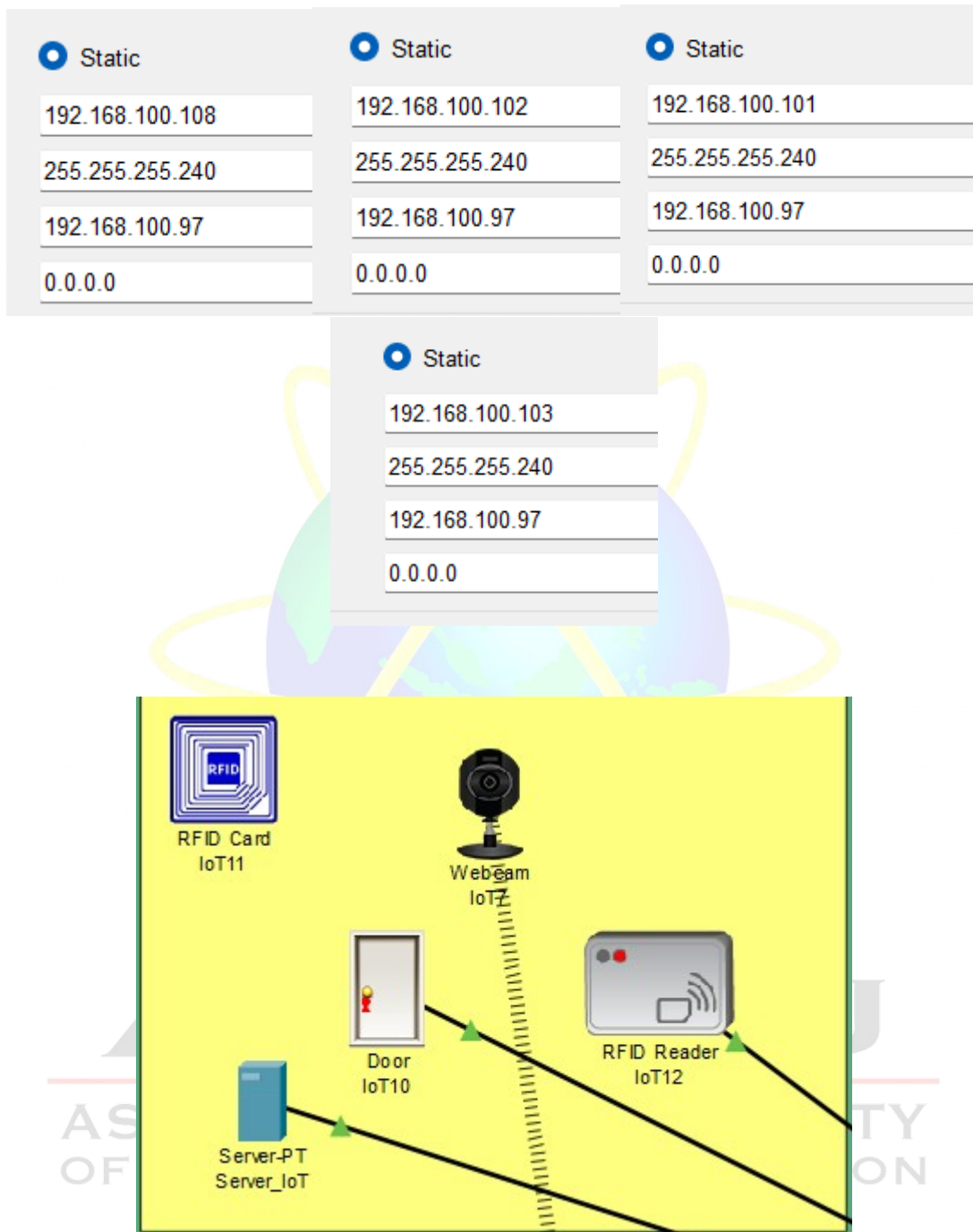


IP address : 192.168.100.104
- 106

And .108

Subnet mask :
255.255.255.224

Default - gateway :
192.168.100.65



Apparatus Stock & Usage

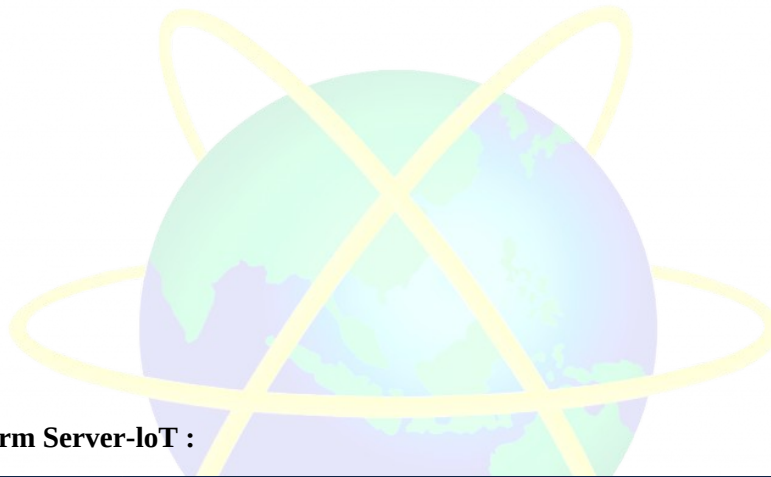
1. Access Control System

- RFID Reader (IoT12): verifies identities
- RFID Card (IoT11): Workforce authorizer

- Automated Door (IoT10): Automated electromechanical entrance point

2. Surveillance System

- Webcam (IoTZ): Camera (Imaging)
- Server Integration
- Server_IoT: Management central of security installation



1. configure form Server-IoT :

Server_IoT 1-

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT**
- VM Management
- Radius EAP

Registration Server

This service runs on top of the HTTP or HTTPS service.

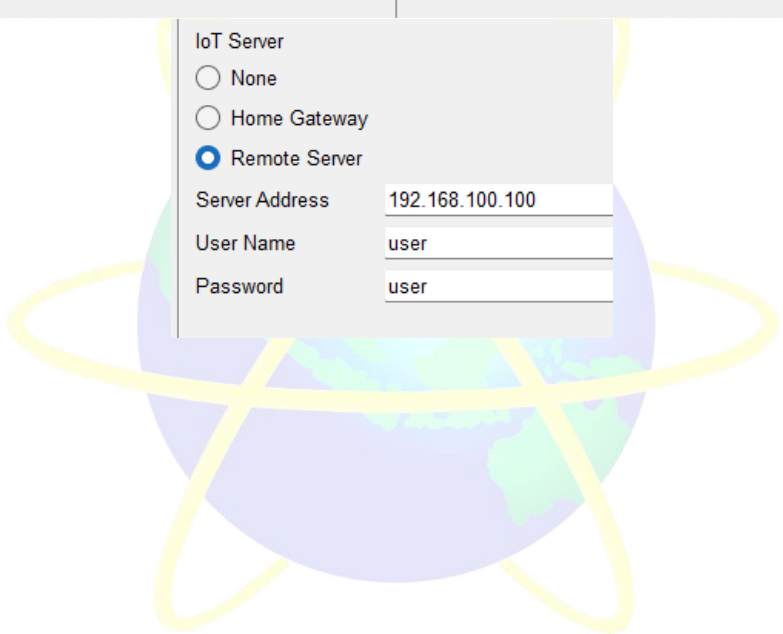
Service ☒ On

	Username	Password
1	user	user

3-

2. Configure from the door , RFID Reader and the webcam :

IoT Server ☐ None ☐ Home Gateway ☒ Remote Server Server Address 192.168.100.100 User Name user Password user	IoT Server ☐ None ☐ Home Gateway ☒ Remote Server Server Address 192.168.100.100 User Name user Password user
---	---



6- Check the end devices are connected to the server :

Server_IoT

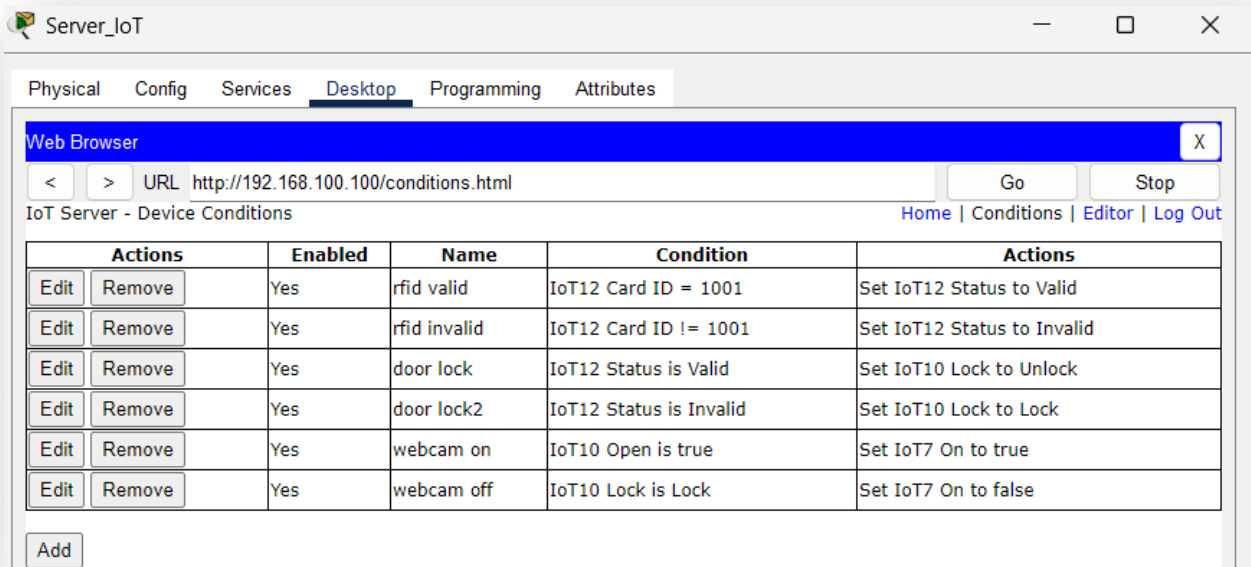
Physical Config Services **Desktop** Programming Attributes

IoT Monitor X

IoT Server - Devices Home | Conditions | Editor | Log Out

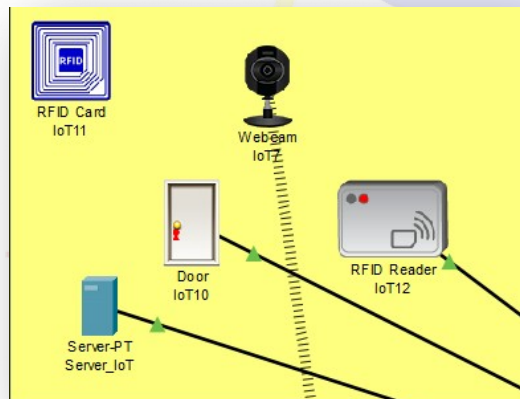
▶ ● IoT12 (PTT08102LB7-)	RFID Reader
▶ ● IoT10 (PTT081015C0-)	Door
▶ ● IoT7 (PTT0810JO82-)	Webcam

7- Add the conditions

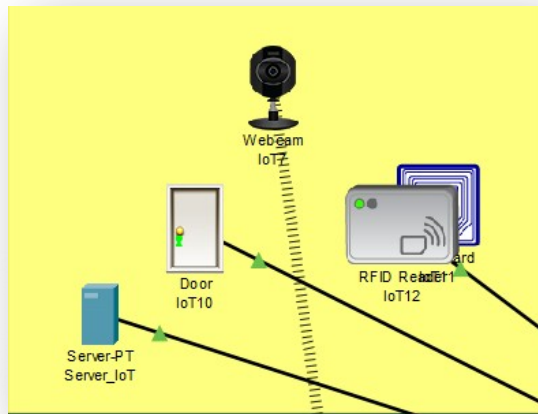


Note : the webcam will turn on only when the door open .

8- Check the conditions if it works :



-- Before --



-- After --

6.0 Subnetting Scheme Table :

Which Subnetting Strategy we have to select ??

We have two options :

1. **Fixed-Length Subnet Mask (FLSM):** All subnets same size
2. **Variable-Length Subnet Mask (VLSM):** Subnets can be different sizes

Why I chose VLSM:

- More efficient use of IP space
- Allows better accommodation of different size requirements
- Leaves more room for future expansion

Subnet Name	No of Hosts	Block Size (2^n)	Subnet Mask	Network Address	Broadcast Address	Usable IP Range
R&D	23	32 (2^5)	255.255.255.224	192.168.100.0	192.168.100.31	192.168.100.1 - 192.168.100.30
Marketing & Sales	19	32	255.255.255.224	192.168.100.32	192.168.100.63	192.168.100.33 -

		(2 ⁵)				192.168.100.62
Technical Support	16	32 (2 ⁵)	255.255.255.224	192.168.100.64	192.168.100.95	192.168.100.65 - 192.168.100.94
Server Room	11	16 (2 ⁴)	255.255.255.240	192.168.100.96	192.168.100.111	192.168.100.97 - 192.168.100.110
Router Link 1	2	4 (2 ²)	255.255.255.252	192.168.100.112	192.168.100.115	192.168.100.113 - 192.168.100.114
Future Expansion	-	32 (2 ⁵)	255.255.255.224	192.168.100.120	192.168.100.151	192.168.100.121 - 192.168.100.150

1. R&D (20 hosts):

- o $2^5 = 32 \rightarrow 32 - 2 = 30$ usable IPs (meets 20 requirement with 10 extra)
- o Next size down ($2^4 = 16$) would only give 14 usable IPs (too small)

2. Marketing (20 hosts):

- o Same as R&D $\rightarrow$ needs /27 (32 IP block)

3. Technical Support (17 hosts):

- o Could use $2^5 = 32$ (30 usable) to match others and allow growth
- o Alternative: $2^4 = 16$ (14 usable) would barely meet current needs but leave no room for growth.

4. Server Room (11 hosts):

- o $2^4 = 16 \rightarrow 16 - 2 = 14$ usable IPs (meets 11 with 3 extra)
- o Next size down ($2^3 = 8$) would only give 6 usable IPs (too small)

5. Router Links:

- o Each link needs exactly 2 usable IPs (one for each router interface)
- o $2^2 = 4 \rightarrow 4 - 2 = 2$ usable IPs (perfect fit)

6- Future Expansion

- Left one /27 block (32 IPs):
 - o Network: 192.168.100.120
 - o Broadcast: 192.168.100.151
 - o Could use $2^5 = 32$ (30 usable).

7.0 Testing & commissioning :

7.1 D&R Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	Laptop11	ICMP		0.000	N	0	(edit)	
	Successful	Lapto...	Laptop15	ICMP		0.000	N	1	(edit)	
	Successful	PC11	Laptop19	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Lapto...	PC15	ICMP		0.000	N	3	(edit)	
	Successful	PC12	Router2	ICMP		0.000	N	4	(edit)	
	Successful	Printer5	Laptop5	ICMP		0.000	N	5	(edit)	

```
C:\>ping 192.168.100.100

Pinging 192.168.100.100 with 32 bytes of data:

Reply from 192.168.100.100: bytes=32 time=10ms TTL=127
Reply from 192.168.100.100: bytes=32 time=22ms TTL=127
Reply from 192.168.100.100: bytes=32 time=9ms TTL=127
Reply from 192.168.100.100: bytes=32 time=40ms TTL=127

Ping statistics for 192.168.100.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 40ms, Average = 20ms

C:\>
```

```
C:\>ping 192.168.100.73

Pinging 192.168.100.73 with 32 bytes of data:

Reply from 192.168.100.73: bytes=32 time=35ms TTL=127
Reply from 192.168.100.73: bytes=32 time=52ms TTL=127
Reply from 192.168.100.73: bytes=32 time=24ms TTL=127
Reply from 192.168.100.73: bytes=32 time=6ms TTL=127

Ping statistics for 192.168.100.73:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 52ms, Average = 29ms

C:\>
```

7.2 Marketing & Sales Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Lapto...	PC9	ICMP		0.000	N	0	(edit)	
	Successful	Lapto...	Server5	ICMP		0.000	N	1	(edit)	
	Successful	PC21	Laptop6	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Lapto...	Laptop20	ICMP		0.000	N	0	(edit)	
	Successful	PC20	Switch1	ICMP		0.000	N	1	(edit)	
	Successful	IoT16	Router2	ICMP		0.000	N	2	(edit)	

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.36

Pinging 192.168.100.36 with 32 bytes of data:

Reply from 192.168.100.36: bytes=32 time<1ms TTL=128
Reply from 192.168.100.36: bytes=32 time=1ms TTL=128
Reply from 192.168.100.36: bytes=32 time<1ms TTL=128
Reply from 192.168.100.36: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

```

C:\>ping 192.168.100.76

Pinging 192.168.100.76 with 32 bytes of data:

Reply from 192.168.100.76: bytes=32 time<1ms TTL=127
Reply from 192.168.100.76: bytes=32 time=7ms TTL=127
Reply from 192.168.100.76: bytes=32 time<1ms TTL=127
Reply from 192.168.100.76: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.100.76:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>

```

7.3 Technical Support Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Server2	Laptop4	ICMP		0.000	N	0	(edit)	
	Successful	PC7	PC6	ICMP		0.000	N	1	(edit)	
	Successful	Printer2	Switch3	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC8	PC14	ICMP		0.000	N	0	(edit)	
	Successful	Laptop6	IoT2	ICMP		0.000	N	1	(edit)	
	Successful	IoT1	IoT4	ICMP		0.000	N	2	(edit)	

```
C:\>ping 192.168.100.99

Pinging 192.168.100.99 with 32 bytes of data:

Reply from 192.168.100.99: bytes=32 time<1ms TTL=254
Reply from 192.168.100.99: bytes=32 time<1ms TTL=254
Reply from 192.168.100.99: bytes=32 time=1ms TTL=254
Reply from 192.168.100.99: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.100.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
C:\>ping 192.168.100.40

Pinging 192.168.100.40 with 32 bytes of data:

Reply from 192.168.100.40: bytes=32 time=5ms TTL=127
Reply from 192.168.100.40: bytes=32 time=7ms TTL=127
Reply from 192.168.100.40: bytes=32 time=4ms TTL=127
Reply from 192.168.100.40: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.100.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 8ms, Average = 6ms

C:\>
```

7.4 Server Room Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC15	Server6	ICMP		0.000	N	0	(edit)	
	Successful	Serve...	IoT7	ICMP		0.000	N	1	(edit)	
	Successful	IoT12	PC14	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Serve...	PC12	ICMP		0.000	N	2	(edit)	
	Successful	IoT7	PC11	ICMP		0.000	N	3	(edit)	
	Successful	IoT8	Laptop21	ICMP		0.000	N	4	(edit)	

```
C:\>ping 192.168.100.6

Pinging 192.168.100.6 with 32 bytes of data:

Reply from 192.168.100.6: bytes=32 time=33ms TTL=127
Reply from 192.168.100.6: bytes=32 time=6ms TTL=127
Reply from 192.168.100.6: bytes=32 time=18ms TTL=127
Reply from 192.168.100.6: bytes=32 time=18ms TTL=127

Ping statistics for 192.168.100.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 33ms, Average = 18ms

C:\>
```

```
C:\>ping 192.168.100.65

Pinging 192.168.100.65 with 32 bytes of data:

Reply from 192.168.100.65: bytes=32 time<1ms TTL=255
Reply from 192.168.100.65: bytes=32 time<1ms TTL=255
Reply from 192.168.100.65: bytes=32 time<1ms TTL=255
Reply from 192.168.100.65: bytes=32 time=8ms TTL=255

Ping statistics for 192.168.100.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>|
```

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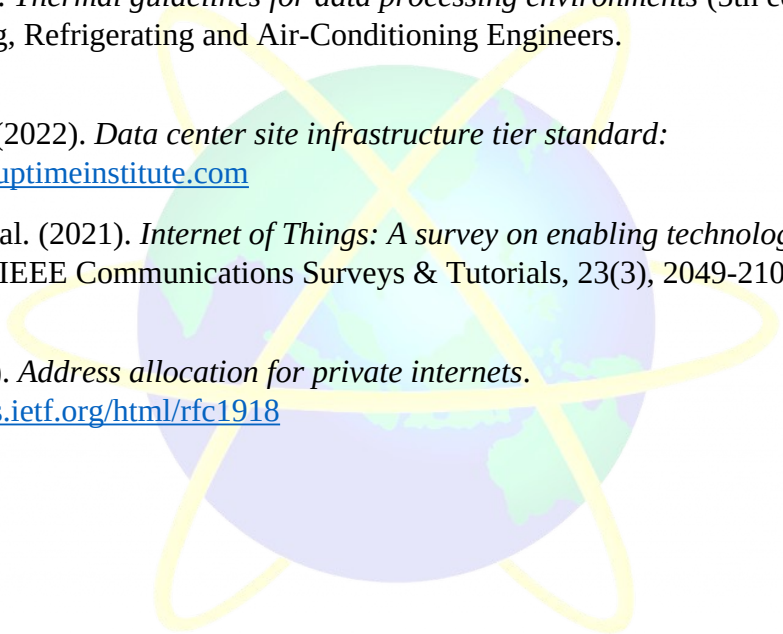
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